

Variational Recovery of Internal Variables in Admissible Spaces: Constrained L^2 Projection for Norm-Like Variables

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Abstract

The variational recovery of internal variables of Mota et al. [1] extends integration-point fields to the entire domain by an L^2 projection derived from a three-field variational principle, and preserves the admissible spaces of group-valued variables by interpolating in the corresponding Lie algebras through the logarithmic and exponential maps. This treatment leaves open the case of norm-like variables such as the equivalent plastic strain, whose admissible set is the closed half-line $[0, \infty)$: the field attains the boundary value zero over finite regions, the half-line is not a group, and in practice the consistent projection produces negative values near steep gradients. This note argues that the correct extension is not a repair of the logarithmic transformation but a change of the admissible set in the same variational principle: minimize the same L^2 distance over the intersection of the finite element space with the convex cone of admissible fields. The resulting constrained projection is the unique solution of a variational inequality; it is idempotent and non-expansive by construction, properties that the unconstrained projection followed by clamping does not possess. Discretely it is a bound-constrained quadratic program on the already-assembled Gram matrix, solvable by a primal–dual active-set method in a small, finite number of iterations. The active set of the constraint is a localized indicator of an under-resolved front and therefore serves directly as an input to error estimation and mesh size-field construction. The note closes with a taxonomy of internal variables by admissible set and an outline of an implementation in Norm.

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1 Background and problem statement

The recovery procedure of [1] arises from a three-field functional in the deformation mapping φ , a *target* internal variable field $\bar{\mathbf{z}}$ defined over the whole domain, and a Lagrange multiplier $\bar{\mathbf{y}}$ that enforces equality between the target field and the *source* field $\mathbf{z}$ available at integration points. Discretization with interpolation functions λ_α for the target field yields the consistent L^2 projection

$$\bar{\mathbf{z}}_h(\mathbf{X}) = \lambda_\alpha(\mathbf{X}) \bar{\mathbf{z}}_\alpha, \quad \bar{\mathbf{z}}_\alpha = \mathbf{H}_{\alpha\beta}^{-1} \boldsymbol{\Sigma}_\beta, \quad \mathbf{H}_{\alpha\beta} := \int_B \lambda_\alpha \lambda_\beta \mathbf{I} \, dV, \quad \boldsymbol{\Sigma}_\beta := \int_B \lambda_\beta \mathbf{z} \, dV, \quad (1)$$

with $\mathbf{I}$ the $q \times q$ identity and q the number of scalar parameters that represent the internal variable. The projection (1) is the minimizer of $\frac{1}{2} \|\bar{\mathbf{z}}_h - \mathbf{z}\|_{L^2(B)}^2$ over the finite element space V_h . The Gram matrix $\mathbf{H}$ is symmetric positive definite, sparse, and of narrow band. Because the projection is a minimizer, it is optimal; because it is a projection onto a subspace, it is idempotent, which is the property that makes it suitable for repeated remap: lumped (diagonalized) variants are not idempotent and diffuse the field under repeated application.

For internal variables that do not live in a linear space, [1] interpolates in the Lie algebra: the field $\mathbf{Z}$ with values in a Lie group G is transformed by the logarithmic map to $\mathbf{z} = \log \mathbf{Z}$ in the Lie algebra $\mathfrak{g}$, which is a vector space; the projection (1) is applied there; and the result is returned to the group by the exponential map. This guarantees, for example, that projected rotations remain rotations, that isochoric deformations remain isochoric, and that a volumetric Jacobian remains positive.

The open question addressed here is the treatment of *norm-like* variables, of which the equivalent plastic strain $\varepsilon^p = \int \dot{\varepsilon}^p \, dt \geq 0$ is the canonical example. Its admissible set is the closed half-line $[0, \infty)$. In practice, the consistent projection of such a variable produces negative values near steep gradients — the projection is a (near-)best L^2 approximation, and best approximations of fields with sharp fronts oscillate. Negative equivalent plastic strain subsequently enters constitutive updates that assume non-negativity, with consequences ranging from silent inaccuracy to failure of the local return mapping.

2 Why the logarithmic transformation does not extend to this case

It is tempting to treat $[0, \infty)$ by analogy with the positive reals $\mathbb{R}^+$, which form a Lie group under multiplication with Lie algebra $\mathbb{R}$. The analogy fails for a structural reason, and identifying that reason yields a criterion that predicts which variables need which treatment.

The half-line $[0, \infty)$ is not a group: zero has no multiplicative inverse, and $\log 0 = -\infty$. More importantly, zero is not an exceptional value of the equivalent plastic strain but its *generic* value: every elastic region carries $\varepsilon^p = 0$ exactly, over finite subdomains. The field therefore sits *on* the boundary of its admissible set, not merely near it.

Contrast this with the variables the logarithmic transformation serves well. The volumetric Jacobian $J = \det \mathbf{F}$ is positive, and $J \rightarrow 0$ is approached only asymptotically under extreme compression; the generic state $J = 1$ maps to $\log J = 0$, the identity of the algebra. Scalar damage $D \in [0, 1)$, transformed as $1 - D \in \mathbb{R}^+$ with $h = \log(1 - D)$, maps the generic virgin state $D = 0$ to $h = 0$; the boundary $D \rightarrow 1$ is asymptotic and is pushed to $h \rightarrow -\infty$, where linear interpolation cannot overshoot it.

The criterion is then:

Asymptotic boundaries — approached but never attained — are handled by a logarithmic transformation, which maps them to infinity. Attained boundaries — on which the field sits over finite regions — cannot be transformed away and must be enforced as inequality constraints.

The two mechanisms compose. Scalar damage possesses both kinds of boundary: the asymptotic boundary $D \rightarrow 1$ is removed by the transformation $h = \log(1 - D) \in (-\infty, 0]$, while the attained boundary $D = 0$, that is $h = 0$, persists in the transformed variable and requires the one-sided constraint $h \leq 0$ — which is again a convex cone constraint, now linear in the transformed space. The equivalent plastic strain has only the attained boundary and is treated by a cone constraint in physical space directly.

A regularized transformation $\log(\epsilon + \varepsilon^p)$ is not a remedy. First, the reconstructed location of an elastic–plastic front becomes a function of the regularization parameter ϵ rather than of the data. Second, the L^2 norm in the transformed space weights *relative* error: fluctuations of order $\log \epsilon$ in the elastic region contribute to the minimized distance as strongly as the physically meaningful variation in the plastic zone, and the minimizer approximates the former in preference to the latter.

3 Constrained projection as a variational inequality

The proposed extension retains the variational principle of [1] verbatim and changes only the admissible set. Let $K \subseteq \mathbb{R}^q$ be the closed convex set of admissible values of the internal variable (a cone, a box, or a simplex; see Section 6), and define the admissible subset of the finite element space

$$\mathcal{K}_h := \{\boldsymbol{\eta}_h \in V_h : \boldsymbol{\eta}_h(\mathbf{X}) \in K \text{ for all } \mathbf{X} \in B\}, \quad (2)$$

which is closed and convex. The constrained recovery is

$$\bar{\mathbf{z}}_h = \arg \min_{\boldsymbol{\eta}_h \in \mathcal{K}_h} \frac{1}{2} \|\boldsymbol{\eta}_h - \mathbf{z}\|_{L^2(B)}^2, \quad (3)$$

equivalently the solution of the variational inequality

$$\bar{\mathbf{z}}_h \in \mathcal{K}_h : \quad (\bar{\mathbf{z}}_h - \mathbf{z}, \boldsymbol{\eta}_h - \bar{\mathbf{z}}_h)_{L^2(B)} \geq 0 \quad \forall \boldsymbol{\eta}_h \in \mathcal{K}_h, \quad (4)$$

the discrete analogue of the obstacle problem. When no constraint is active, (3) reduces identically to (1); on a mesh that resolves the field, the constrained and unconstrained recoveries coincide.

The essential properties follow from the Hilbert space projection theorem onto closed convex sets.

Proposition 1 (Well-posedness, idempotency, stability). *Problem (3) admits a unique solution $\bar{\mathbf{z}}_h = P_{\mathcal{K}_h} \mathbf{z}$. The map $P_{\mathcal{K}_h}$ is idempotent, $P_{\mathcal{K}_h} \circ P_{\mathcal{K}_h} = P_{\mathcal{K}_h}$, and non-expansive, $\|P_{\mathcal{K}_h} \mathbf{u} - P_{\mathcal{K}_h} \mathbf{v}\| \leq \|\mathbf{u} - \mathbf{v}\|$.*

Idempotency deserves emphasis because it has been raised as a concern: one might fear that appending constraints to the projection destroys the idempotency that motivates the consistent projection in the first place. The opposite holds. The metric projection onto a closed convex set is idempotent *by construction* — an admissible field is its own closest admissible point — whereas the apparently simpler alternative, the unconstrained projection followed by a pointwise clamp, is not: the clamp is the projection in the wrong (diagonal) metric, the clamped field is generally not the L^2 -closest admissible field, and re-projecting it re-introduces oscillation. Under the repeated project–interpolate cycles of successive remaps, the composition “project, then clamp” drifts; the constrained projection does not. Non-expansiveness supplies the complementary stability statement: a sequence of constrained projections cannot amplify perturbations.

Remark 1 (One-sided conservation of the accumulated quantity). *Let $K = [0, \infty)^q$, componentwise non-negativity of the q scalar parameters that represent the internal variable in (1), with the equivalent plastic strain as the scalar case $q = 1$. The unconstrained projection conserves the integral of the field: the Lagrange interpolation functions form a partition of unity, $\sum_\alpha \lambda_\alpha = 1$, so constant fields belong to V_h , and testing the unconstrained stationarity condition $(\bar{\mathbf{z}}_h - \mathbf{z}, \boldsymbol{\eta}_h)_{L^2(B)} = 0$ with $\boldsymbol{\eta}_h = \mathbf{1}$, the constant field of ones in $\mathbb{R}^q$, yields $\int_B \mathbf{1} \cdot \bar{\mathbf{z}}_h \, dV = \int_B \mathbf{1} \cdot \mathbf{z} \, dV$.*

The constrained projection retains exactly one half of this statement, and the geometry of the cone selects which half. In the variational inequality (4) the perturbation $\boldsymbol{\eta}_h = \bar{\mathbf{z}}_h + c\mathbf{1}$, with $c > 0$ a scalar constant, is always admissible — raising an admissible field keeps it admissible — and substitution gives $c \int_B (\bar{\mathbf{z}}_h - \mathbf{z}) \cdot \mathbf{1} \, dV \geq 0$, that is,

$$\int_B \mathbf{1} \cdot \bar{\mathbf{z}}_h \, dV \geq \int_B \mathbf{1} \cdot \mathbf{z} \, dV. \quad (5)$$

The opposite perturbation $\boldsymbol{\eta}_h = \bar{\mathbf{z}}_h - c\mathbf{1}$ is inadmissible wherever any component of the solution touches the bound, so no matching upper inequality survives: the projection may raise the total of the field, never lower it.

Discretely the statement sharpens to an identity. In the scalar-variable setting of Section 4, multiplying the complementarity system (8) by $\mathbf{1}^\top$, with $\mathbf{1}$ the vector of ones, and using the partition of unity twice, namely $\mathbf{1}^\top \mathbf{H} \bar{\mathbf{z}} = \int_B \bar{z}_h \, dV$ and $\mathbf{1}^\top \boldsymbol{\Sigma} = \int_B \mathbf{z} \, dV$, gives

$$\int_B \bar{z}_h \, dV - \int_B \mathbf{z} \, dV = \mathbf{1}^\top \boldsymbol{\mu} \geq 0, \quad (6)$$

so the excess of the accumulated variable equals the resultant of the admissibility reactions $\boldsymbol{\mu}$ of Remark 2. It is therefore supported on the active set, vanishes identically when no constraint activates, and decays with the local projection error as the mesh resolves the front.

The direction of inequality (5) is the physically correct one. Repairing a negative undershoot of the equivalent plastic strain must add plastic history in a neighborhood of the front; a repair able to remove it would fabricate less-yielded material out of a numerical oscillation. Among all admissible fields the constrained projection adds the least, since $\bar{\mathbf{z}}_h$ is by definition the closest admissible field to the source.

Remark 2 (Interpretation of the multiplier). *In [1] the Lagrange multiplier of the three-field functional is identified with the thermodynamic force conjugate to the internal variable. Appending the inequality $\bar{\mathbf{z}}_h \in K_h$ introduces a further multiplier $\boldsymbol{\mu} \geq \mathbf{0}$, complementary to the constraint, which plays the role of the reaction that maintains admissibility of the state — the same role the contact pressure plays for the impenetrability constraint. Its support marks precisely the regions where the unconstrained projection would exit the admissible set; Section 7 exploits this.*

4 Discrete formulation and solution algorithms

4.1 Bound-constrained quadratic program

For a scalar variable with $K = [0, \infty)$ and constraints imposed on the vector $\bar{\mathbf{z}}$ of nodal values (the adequacy of nodal constraints is discussed in Section 5), problem (3) is the bound-constrained quadratic program

$$\min_{\bar{\mathbf{z}} \geq \mathbf{0}} \frac{1}{2} \bar{\mathbf{z}}^\top \mathbf{H} \bar{\mathbf{z}} - \boldsymbol{\Sigma}^\top \bar{\mathbf{z}}, \quad (7)$$

with $\mathbf{H}$ and $\boldsymbol{\Sigma}$ exactly as assembled for (1). The first-order conditions are the linear complementarity system

$$\mathbf{H} \bar{\mathbf{z}} - \boldsymbol{\Sigma} = \boldsymbol{\mu}, \quad \bar{\mathbf{z}} \geq \mathbf{0}, \quad \boldsymbol{\mu} \geq \mathbf{0}, \quad \boldsymbol{\mu}^\top \bar{\mathbf{z}} = 0. \quad (8)$$

Because $\mathbf{H}$ is symmetric positive definite, (7) is strictly convex and possesses exactly one solution: the closest admissible field of (3). The two algorithms below compute this same point and differ only in how they reach it — the first exactly, in finitely many steps; the second in the limit of an infinite iteration.

4.2 Primal–dual active set (preferred for a new implementation)

The primal–dual active-set method — equivalent to a semismooth Newton method on the complementarity system [2] — terminates at the exact solution of (7) in a finite number of iterations, typically fewer than five in practice. Each iteration fixes the currently active nodes at the bound and solves the reduced symmetric positive definite system on the inactive set. A factorization of $\mathbf{H}$, computed once per remap, is reused across iterations, so the additional computation relative to the unconstrained projection (1) amounts to a small number of back-substitutions on subsets of the unknowns. When no constraints activate, the method terminates after one iteration with the unconstrained solution.

Two further properties recommend it in the remap setting. First, it warm-starts: the front moves little between successive remaps, so the active set of the previous remap is an excellent initial guess, and the method typically confirms or barely amends it in one or two iterations. Second, it yields the multiplier $\boldsymbol{\mu}$ exactly at termination, and the multiplier is precisely the quantity that Section 7 supplies to the size field: a truncated iterative method produces $\boldsymbol{\mu}$ only to the accuracy of its stopping tolerance, contaminating the small, localized indicator one wishes to read.

4.3 Dykstra’s alternating projections (preferred when retrofitting a legacy code)

An implementation that prefers to compose existing operators rather than introduce a quadratic programming kernel may use Dykstra’s algorithm [3]: alternate the consistent projection onto V_h (a Gram matrix solve) and the pointwise clamp onto the cone of non-negative fields, with Dykstra’s correction increments carried between passes. The correction increments are essential, and their role is the point most easily misread. *Plain* alternating projections — project, clamp, repeat, with no corrections — converge to *some* point of the intersection of V_h with the cone: an admissible field, but in general not the minimizer of (7), hence a suboptimal recovery. Dykstra’s increments repair exactly this defect: the corrected iteration converges to the solution of (7), the same point the active-set method computes. The difference between the two subsections is therefore not the answer but the route: the active-set method reaches it exactly in finitely many steps, whereas Dykstra’s iteration reaches it only in the limit, at a linear rate that degrades precisely where the constraint matters — near a sharp front, where the subspace and the cone meet at a shallow angle.

The advantage of Dykstra’s algorithm is therefore architectural rather than mathematical: it introduces no new solver kernel, only the composition of two operators a legacy code already possesses — the consistent Gram matrix solve and the pointwise clamp. In an existing code where a bound-constrained solver would be an intrusive addition, it is the appropriate first implementation; in a new implementation, where the active-set solver of Section 4.2 is straightforward to write, nothing else favors it.

4.4 Quadrature consistency

The unconstrained projection is idempotent under inexact quadrature provided $\mathbf{H}$ and $\mathbf{\Sigma}$ are integrated with the same rule: the projection of a member of V_h then reproduces it identically. The constrained projection inherits this property, since an admissible member of V_h satisfies (4) trivially. The discipline of [1] — the same integration points for the recovery as for the equilibrium statement — carries over unchanged.

5 Where to impose the bounds

Nodal constraints $\bar{\mathbf{z}}_\alpha \geq \mathbf{0}$ guarantee a non-negative *field* only when the interpolation functions are non-negative, as for linear simplices and multilinear hexahedra. For quadratic and serendipity elements the basis functions change sign and nodal bounds do not imply field bounds. Two options exist:

1. constrain the field at the points where it will be evaluated — the integration points of the target mesh — which is natural for in-core remap where the target is known at projection time; or
2. express the element polynomial in the Bernstein basis and constrain the Bernstein coefficients, whose non-negativity is a sufficient (mildly conservative) condition for non-negativity of the polynomial. This is the standard device of bound-preserving high-order transport and requires no knowledge of the target mesh, which suits file-based remap.

For the linear elements that dominate current practice, nodal constraints suffice.

6 A taxonomy of internal variables

Table 1 classifies common internal variables by admissible set, boundary character, and the resulting treatment. The classification is the operational content of the criterion of Section 2.

Variable	Admissible set	Boundary character	Treatment
Back stress, strains	vector space	none	projection (1)
Rotations $\mathbf{R}$	$SO(3)$	none (group)	log/exp + (1)
Isochoric $\mathbf{F}^p$	$SL(3)$	none (group)	log/exp + (1)
Jacobian J	$(0, \infty)$	0 asymptotic	log + (1)
Equiv. plastic strain ε^p	$[0, \infty)$	0 attained	cone constraint (3)
Scalar damage D , porosity f	$[0, 1)$	0 attained, 1 asymptotic	$h = \log(1 - D)$, constraint $h \leq 0$
Volume fractions ϕ_i	simplex	faces attained	(3) with $\sum_i \phi_i = 1$

Table 1: Internal variables by admissible set. “Attained” boundaries are those on which the field sits over finite regions; they require inequality constraints. “Asymptotic” boundaries are removed by logarithmic transformation. The two mechanisms compose, as in the damage row.

The architectural consequence is that the admissible set is *metadata of the state variable*, declared by the constitutive model and read by the transfer machinery, which dispatches on it: plain projection for vector-space variables, logarithmic transformation for group-valued variables, constrained projection for cone- and box-valued variables, and their composition where the taxonomy requires it. Codes that already tag variables for transfer (for instance, as eligible for “ L^2 projection”) possess the natural place for this declaration; the tag needs only to be enriched from a method label to an admissible-set descriptor.

A complementary safeguard belongs in the constitutive model itself: state entering a material update — from initial conditions, restart, or transfer — should be checked against the declared admissible set, with tolerance semantics: violations below a tolerance are repaired silently, violations above it are reported as errors. This is a verification layer, not the primary mechanism. A silent clamp buried in the material model conceals transfer defects and re-introduces exactly the non-idempotent repair that Section 3 argues against; if the transfer enforces admissibility variationally, the material-level check should never activate.

7 Interaction with error estimation and mesh adaptation

The negativity that motivates this note is a symptom; the underlying cause is a front that the mesh does not resolve. The constrained projection removes the inadmissibility but not the approximation error, and it is therefore a complement to, not a substitute for, adaptive refinement driven by the recovery error. The two interact favorably in both directions.

First, the multiplier μ of (8), obtained as a byproduct of the solution, is a localized indicator of exactly the regions where the finite element space cannot represent the front: its support is the active set, which is nonempty precisely where the unconstrained projection would exit the admissible set. In a remap-driven adaptive cycle — project the field, interpolate back to the integration points, measure the L^2 discrepancy elementwise, construct a size field, remesh — the multiplier supplies an additional element-level input to the size field without further computation, identifying fronts that the plain L^2 discrepancy may under-weight because the offending oscillations are small in norm while large in consequence.

Second, as refinement resolves the front, the constraints deactivate and the constrained projection reduces continuously to the unconstrained one, whose optimality and idempotency are already established. The constraint governs the pre-asymptotic regime; refinement removes the need for it. Near a genuine free boundary the recovered field remains only C^0 at any resolution, so mild residual constraint activity at the front is expected and benign.

8 Approaches considered and rejected

Clamping after projection. Not idempotent; the projection in the wrong metric; drifts under repeated remap (Section 3).

Regularized logarithm $\log(\epsilon + \varepsilon^p)$. Front location depends on ϵ ; relative-error weighting resolves the elastic region in preference to the plastic zone (Section 2).

Lumped projection. Idempotency fails; repeated remap diffuses the field. This is the original motivation for the consistent projection of [1] and is not revisited here.

Conservation constraint on the mean. Appending $\int_B \bar{z}_h dV = \int_B z dV$ to $\mathcal{K}_h$ is convex and implementable, but the equivalent plastic strain is not a conserved quantity; the one-sided inequality (5) is the appropriate statement, and forcing equality would spread the front’s admissibility correction over the domain.

9 Outline of an implementation in Norma

Norma is well suited for a prototype: it already assembles consistent mass (Gram) matrices, carries constitutive state at integration points, and possesses the quadrature and interpolation infrastructure of [1]. A minimal introduction would comprise:

1. a recovery module implementing (1) with the Gram matrix assembled by the existing interpolation and quadrature routines, verified for idempotency under project–interpolate cycles on a fixed mesh;
2. the admissible-set tag on constitutive state (vector space, Lie group with its log/exp pair, cone, box, simplex), with the dispatch of Section 6;
3. the primal–dual active-set solver for (7), exercised first on a manufactured front (a piecewise field with an attained-boundary region) and then on a Taylor-bar-type problem under repeated remap, with the clamped unconstrained projection as the control demonstrating drift;
4. the multiplier-to-size-field hook of Section 7, deferred until a remeshing loop exists.

Items 1–3 are self-contained and testable without any remeshing capability, since idempotency, admissibility, and the front benchmark are all statements about a single mesh.

10 Summary

The variational principle of [1] — recover the internal variable field by minimizing the L^2 distance to the source data in the variable’s own space — already contains the answer for norm-like variables. Their own space is a convex cone rather than a group, so the minimization runs over the admissible convex subset of the finite element space and becomes a variational inequality: a bound-constrained quadratic program on the already-assembled Gram matrix, solved by a primal–dual active-set method with a few additional back-substitutions. Idempotency and stability under repeated remap follow from convexity rather than requiring preservation; the logarithmic transformation retains its role for asymptotic boundaries and composes with the constraints where a variable possesses boundaries of both kinds; and the active set of the constraint serves also as the front indicator that an adaptive remeshing loop requires.

References

- [1] A. Mota, W. Sun, J. T. Ostien, J. W. Foulk III, K. N. Long. Lie-group interpolation and variational recovery for internal variables. *Computational Mechanics* 52 (2013) 1281–1299.
- [2] M. Hintermüller, K. Ito, K. Kunisch. The primal–dual active set strategy as a semismooth Newton method. *SIAM Journal on Optimization* 13 (2003) 865–888.
- [3] J. P. Boyle, R. L. Dykstra. A method for finding projections onto the intersection of convex sets in Hilbert spaces. In: *Advances in Order Restricted Statistical Inference*, Lecture Notes in Statistics 37, Springer, 1986, 28–47.
- [4] C. L. Lawson, R. J. Hanson. *Solving Least Squares Problems*. Prentice-Hall, 1974.